

Heart-Rate Data Processing - Part 1

You are a junior data engineer at a Taiwan-based wearable health-tech firm called Seng-Links. Your company has just discovered that the raw heart-rate exports from prototype devices are inconsistent. Some files contains malformed records, impossible values, or missing entries. Before the machine learning team can study usage and identify sleep & exercise patterns, the data engineering team must build a batch-processing pipeline that reads raw files cleans records, computes quality checks, and calculates descriptive statistics for later analysis.

Your company has provided you a folder named "data/", which contains 4 files of heart-rate samples from one participant. The participants device records heart rate every 5 minutes. Your task is to write a Python program that:

1. Reads raw files from the `data/` folder.
2. Cleans the records by ignoring invalid or missing values.
3. Computes and prints descriptive statistics for each file.

Learning Objectives

- Use **Python file I/O** to read in binary data for data processing and use **typecasting** to convert strings into integer data-types.
- Use **for-loops** to calculate the **average** of a Python list and detect malformed data via **conditionals**.
- Document complex chunks of code using **Python comments** and introductory **docstrings**.
- Create **functions** to begin to understand how modules enable separation of responsibilities and integrating separate components to support systems-level thinking.
- **(advanced)** Calculate **rolling-average** without using the **statistics** package in Python. Aim for a computationally efficient solution (no double-nested for-loops).

Instructions

Download this [folder](#) **Download folder** to get started with your project.

Begin this project by reading all sections of the **instructions.md** document. Once you understand the directions, utilize documentation, your peers, readings, and classroom notes to complete this project. Some sections of this project have been commented for you, so your task will entail interpreting the comments in order to write the intended line of code. Other sections will test your ability to **independently** implement code with minimal guidance.

Resources

Some relevant internal resources to help you complete this project include the following links:

- **Intro to TLAB 1 (3/12/2026)**

- https://drive.google.com/file/d/1Ew0M_s-dKdJ6-FVnsLA4TwlDunbGDydo/view?usp=sharingLinks to an external site.
- **GitHub Review Session (3/20/2025)**
 - https://theknowledgehouse-org.zoom.us/rec/share/3G7NOqdJGTka4pSCJ0pQU1b1NhV1Js8Nj3_460R9G8glU5Gh1mzHgSMHy_0hARPf.jyWPzZPYwnGHgW6S?startTime=1742505914000Links to an external site.
 - **Passcode:** S#7Hwqcv

Submission

This TLAB will be due 3/20. You have two options to submit your assignment:

1. Submit a link to your Google drive containing your complete code.
2. Submit a link to your GitHub repository containing your completed code

You have 2 attempts for this project. If you submit your project at least 1 week before the due date (3/13) , you will be able to receive an early grade. You can use feedback from this grade to implement any corrections and subsequently resubmit for an opportunity for a higher grade.